

IT Help Desk Chatbot: Part 2

Measuring and improving retrieval. Repo: github.com/mithransadasivam/FDE, folder "Chatbot project", tag v2.0-day7.

What Part 2 was about

In Part 1 the chatbot worked, but I never measured how good its search was. In Part 2 I made a 20-question test set, measured the search, tried three improvements (hybrid search, reranking and query rewriting), compared all four ways of searching, and picked one using my own numbers.

My target, set before I measured anything: the right chunk is in the top 4 for at least 90% of the questions, and a whole answer takes 3 seconds or less. If more than one mode met the target, I would keep the simplest.

The test set

20 questions in **data/retrieval_test_set.csv**: 6 plain, 4 reworded, 4 with exact codes, 3 follow-ups and 3 messy ones. Each question lists the document the answer should come from and a short phrase that must appear in the right chunk. The knowledge base has 23 documents (62 chunks): my 17 Northwind documents plus the 6 class PDFs.

The evidence check confirms every phrase really exists in its document:

Evidence check: 20 of 20 found

The four modes

Mode	What it does
vector	Finds chunks with similar meaning (what Part 1 used)
hybrid	Vector search plus keyword search, then the two lists are merged
rerank	Hybrid, then the model re-orders the best 10 candidates
full	Rewrites a follow-up question using the previous one, then hybrid, then rerank

Results

Mode	Hit rate (top 4)	MRR	Search time	Model calls per answer	Whole answer	Answer-or-decline test	Meets target
vector (chosen)	95%	0.79	123 ms	1	1.97 s	10/10	yes
hybrid	90%	0.87	111 ms	1	2.11 s	10/10	yes
rerank	100%	0.94	2272 ms	2	4.01 s	10/10	no
full	95%	0.91	3411 ms	3	5.63 s	10/10	no

Hit rate = how often the right chunk is in the top 4. MRR = how high up the right chunk is on average (closer to 1 is better).

My choice: vector

Plain vector search already meets both parts of my target, so I kept it. Hybrid also meets the target but found fewer right chunks. Rerank was the most accurate but is too slow, and full was slower still without being better.

I ran my Part 1 behaviour test again with vector (saved as run 3): **10 out of 10 correct**, 7 of 7 answered, 3 of 3 declined, 2.13 seconds on average. All four modes passed this test, so a better search did not break the chatbot's habit of declining questions it can't answer. The full reasoning is in RAG_DECISION.md.

Screenshots

Retrieval tab (vector chosen; the side-by-side view compares vector and full on "And on a Mac?"):

IT Help Desk Assistant

Answers only from your team's documents

KNOWLEDGE BASE

Northwind IT docs READY

23 documents · 62 chunks

SETTINGS

Setting	Value
Chunk size	500
Overlap	100
Chunks retrieved	4
Minimum score	0.55

RETRIEVAL MODE

vector

hybridrerankfull

Setting	Value
Candidates reranked	-
Chunks in the prompt	4
Model calls per question	0 + 1

RETRIEVAL TEST SET

Item	Value
File	retrieval_test_set.csv
Questions	20
Evidence phrases found	20 / 20
Kinds (plain · reworded · code · follow-up · messy)	6 · 4 · 4 · 3 · 3

DECISION

Item	Value
Target	≥ 90% · under 3 s
Chosen mode	vector
Recorded in	RAG_DECISION.md

IT Help Desk Assistant

Answers only from your team's documents · every fact cited

Chat

Knowledge base

Test results

Retrieval

20 questions · 4 modes · saved to data/retrieval_comparison.csv

Hit rate @4 (vector)

MRR (vector)

Hit rate gained over vector

Time per answer

95%

0.79

+0 pts

2.0 s

Mode	Hit @4	MRR	plain	reworded	exact code	follow-up	messy	Retrieval ms	Model calls	Answers	Answer or decline
✓ vector	95%	0.79	6/6	4/4	4/4	2/3	3/3		123 0 + 1	2.0	10/10
hybrid	90%	0.87	6/6	3/4	4/4	3/3	2/3		111 0 + 1	2.1	10/10
rerank	100%	0.94	6/6	4/4	4/4	3/3	3/3		2272 1 + 1	4.0	10/10
full	95%	0.91	6/6	4/4	4/4	2/3	3/3		3411 2 + 1	5.6	10/10

Target (set before measuring): hit rate at least 90% and a whole answer within 3 s. Model calls: retrieval + the call that writes the answer.

What was found: two modes side by side

Question

[follow-up] And on a Mac?

Mode A

vector

What was found · vector — right chunk at rank 2 (hit)

Searched for: And on a Mac?

· wifi_guide.md p.1

✓ vpn_setup_guide.md p.1

✓ vpn_setup_guide.md p.1

· Day06_Slide15_policy_vpn_remote_access.pdf p.1

✓ marks the expected document: vpn_setup_guide.md. Evidence: "Approve the system extension"

Mode B

full

What was found · full — right chunk at rank 1 (hit)

Searched for: How do I install the VPN client on Mac

✓ vpn_setup_guide.md p.1

✓ vpn_setup_guide.md p.1

· Day07_Slide05_kb_error_codes.pdf p.1

· Day07_Slide05_kb_error_codes.pdf p.1

A follow-up question in the chat. "and on a mac?" was searched as "VPN keeps cutting out Mac", shown on the "Searched for" line, and the answer cites the right document. (This one was taken in full mode, because that is the mode that rewrites the question.)

IT Help Desk Assistant

Answers from your team's documents

KNOWLEDGE BASE

Northwind IT docs READY

23 documents · 62 chunks

SETTINGS

Setting	Value
Chunk size	500
Overlap	100
Chunks retrieved	4
Minimum score	0.55

RETRIEVAL MODE

vector hybrid rerank **full**

Setting	Value
Candidates reranked	10
Chunks in the prompt	4
Model calls per question	2 + 1

RETRIEVAL TEST SET

Item	Value
File	retrieval_test_set.csv
Questions	20
Evidence phrases found	20 / 20
Kinds (plain · reworded · code · follow-up · messy)	6 · 4 · 4 · 3 · 3

DECISION

Item	Value
Target	≥ 90% · under 3 s
Chosen mode	full

IT Help Desk Assistant

Answers only from your team's documents · every fact cited

Chat

Knowledge base

Test results

Retrieval

🗨️

My VPN keeps cutting out at home. What can I try?

📖

Here are steps to fix VPN disconnections at home:

1. Check your NorthLink version - Make sure you're running version 5.2 or later, as older versions drop connections on weak Wi-Fi [1]

2. Improve your connection - If you're on Wi-Fi, move closer to your router or connect with a network cable instead [1]

3. Change the protocol - Switch the protocol in NorthLink settings from "Auto" to "TCP". This fixes most random disconnects [1]

4. Note the 12-hour limit - Your VPN session automatically disconnects after 12 hours, so you'll need to reconnect and sign in again [3][4]

If these steps don't resolve the issue, open a ticket on the service desk portal and attach your NorthLink log file (Help > Export logs) [2]

[1] vpn_troubleshooting.md · page 1

[2] vpn_troubleshooting.md · page 1

[3] vpn_setup_guide.md · page 1

[4] Day06_Slide15_policy_vpn_remote_access.pdf · page 1

14.6 s · 4 chunks

Searched for: VPN keeps disconnecting at home troubleshooting

🗨️

and on a mac?

📖

Based on your question about VPN setup on a Mac:

1. Install "NorthLink VPN" [2]

2. Approve the system extension in System Settings > Privacy & Security, then restart the Mac once [2]

3. Start NorthLink, use the same server address, and sign in [2]

Note: Your VPN session will automatically disconnect after 12 hours of connection, and you'll need to reconnect and sign in again [2].

[2] vpn_setup_guide.md · page 1

4.92 s · 4 chunks

Searched for: VPN keeps cutting out Mac

Reflection

Which technique helped the most, and why. Reranking. It was the only mode to get all 20 questions right (100%) and it had the best MRR (0.94, against 0.79 for plain vector search). Plain search often found the right chunk but put it second or third. Reranking asks the model to look at the top 10 and put the best one first. The problem is time: it adds a model call to every question, so a whole answer took about 4 seconds, over my 3 second target. That is why I did not choose it.

One that did not help. Query rewriting (full mode). It does fix follow-ups like "And on a Mac?", but hybrid and reranking had already fixed most of them, so it added almost nothing. It scored 95% and was slower than reranking (5.6 seconds, three model calls per answer). Hybrid was also a disappointment: its hit rate was lower (90%) because chatty words in a question matched the wrong documents, and my exact-code questions already worked with plain vector search.

One question that still fails. With vector, "What if it still fails after that?" still fails. It follows a VPN question, but on its own the words "still fails" say nothing about VPNs, so the right chunk came 5th, outside the top 4. Rewriting with the previous question fixes it, but I decided one question in 20 was not worth the extra time.

Things to know

- 20 questions is not many. One question is 5 points of hit rate, so the gap between vector (95%) and hybrid (90%) is a single question.

- My Northwind documents and the class PDFs disagree on 5 facts (for example password expiry). The bot can quote either one. I did not fix this.
- All files are in the repo: RAG_DECISION.md, PART2_NOTES.md, REQUEST_LOG.md (Part 2 entries 21 to 31), data/retrieval_comparison.csv and the 205 offline tests.